Transferability estimation

Which pre-trained model should I fine-tune, and can I tell without fine-tuning them all?

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The problem · six scores from NCE to PAS · two benchmarks

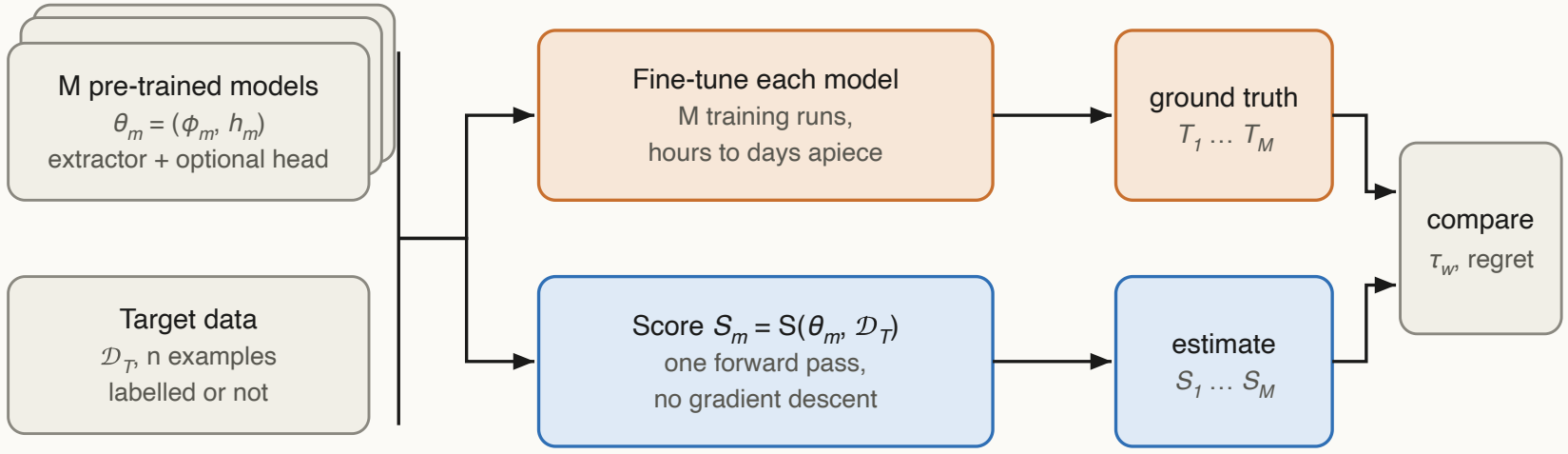
Built from the annotated notes in `theory_inclined_papers_with_annotations`.

The practical question

A model zoo holds M pre-trained models. You have one new target task with n examples.

- **Gold standard:** fine-tune all M, evaluate, keep the best. Cost: M training runs, each with its own hyper-parameter search
- **Transferability estimation:** compute a cheap score S_m per model, from the frozen model and the target data, and take the top one

The problem in one picture: rank the pool without fine-tuning every model



Success means the cheap ranking (blue) reproduces the expensive one (orange).

The problem, in symbols

Given

- a pool $\theta_m = (\phi_m, h_m), m = 1, \dots, M$: feature extractor $\phi_m : \mathcal{X} \to \mathbb{R}^{D_m}$, optional head h_m from a source task $\mathcal{T}_m^S$ (its data $\mathcal{D}_m^S$ may be unavailable)
- target data $\mathcal{D}^T$ of n examples, and a test distribution P^T

Ground truth (expensive): $T_m = \mathrm{perf}\left(\mathrm{FT}(\theta_m; \mathcal{D}^T);\ P^T\right)$, the accuracy, balanced accuracy, AUROC or log-likelihood after fine-tuning (or after retraining only the head)

Estimator (cheap): $S_m = S(\theta_m; \mathcal{D}^T)$, one forward pass, no gradient descent

Goal: make $\tau_w\big((S_m)_m, (T_m)_m\big)$ high, or, for pure selection, make the regret $\max_m T_m - T_{\hat{m}}$ small, where $\hat{m} = \arg\max_m S_m$

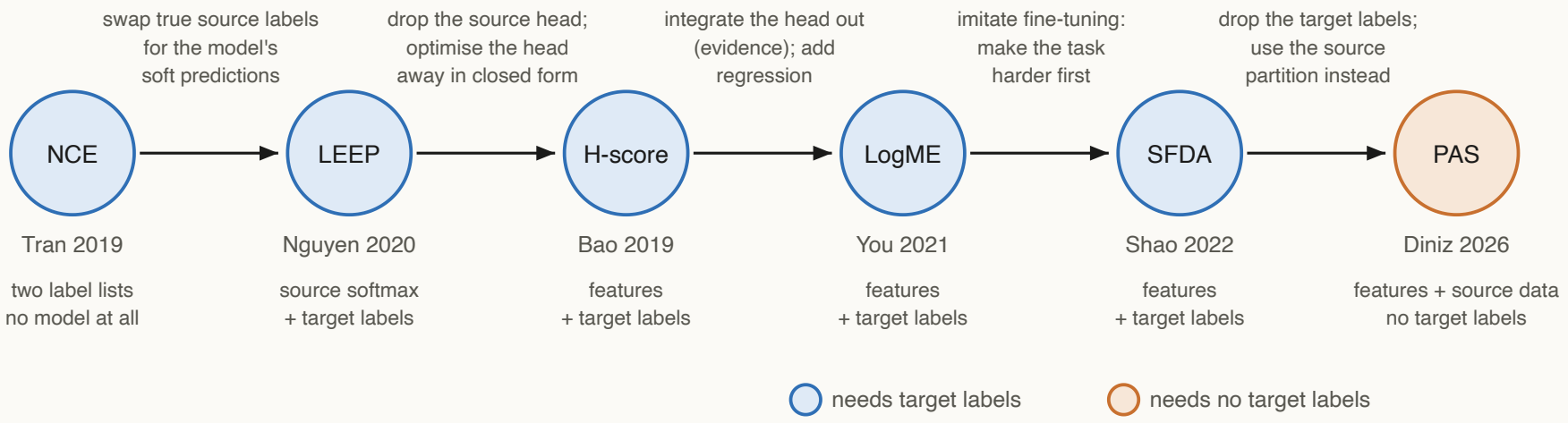
How a score is judged

- Kendall's $\tau = \dfrac{2}{M(M-1)} \sum_{i<j} \operatorname{sgn}(T_i - T_j) \operatorname{sgn}(S_i - S_j)$: pairs ordered the same way minus pairs ordered oppositely
- If the correlation is τ, then $\Pr(T_i > T_j \mid S_i > S_j) = \frac{\tau+1}{2}$, so $\tau_w \approx 0.5$ means about 75% of pairs ordered right (LogME paper)
- **Weighted** τ_w counts a pair more when T_i, T_j, S_i, S_j are large, because you deploy the top-ranked model
- Also used: Spearman ρ, Pearson r, the regret of the top pick

Notes: You 2021 — LogME (the τ_w definition).

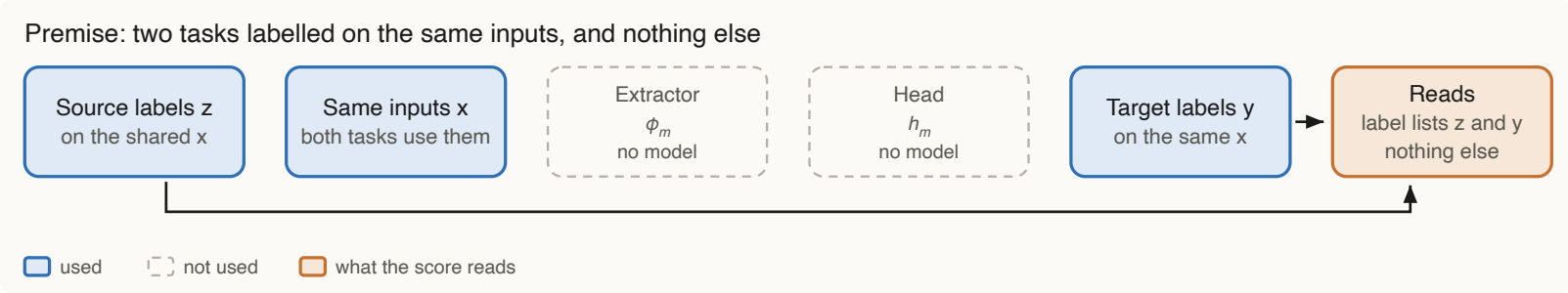
The progression we will follow

From the tractable case to the label-free one: what changes at each step



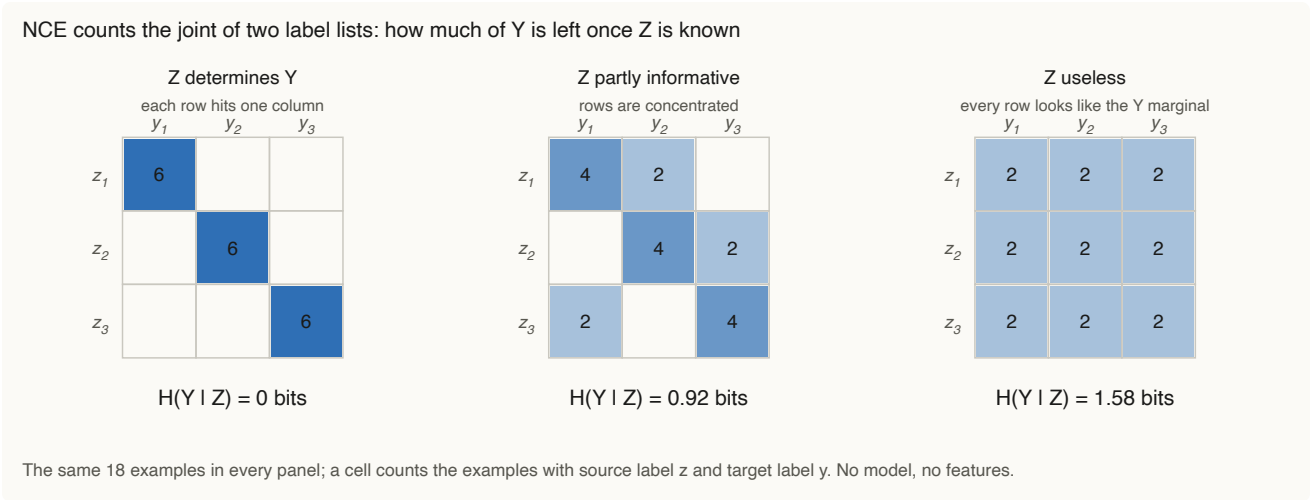
A conceptual order, not a chronological one: NCE and H-score are both 2019, LEEP 2020, LogME 2021, SFDA 2022, PAS 2026.

NCE (Tran 2019): the tractable case

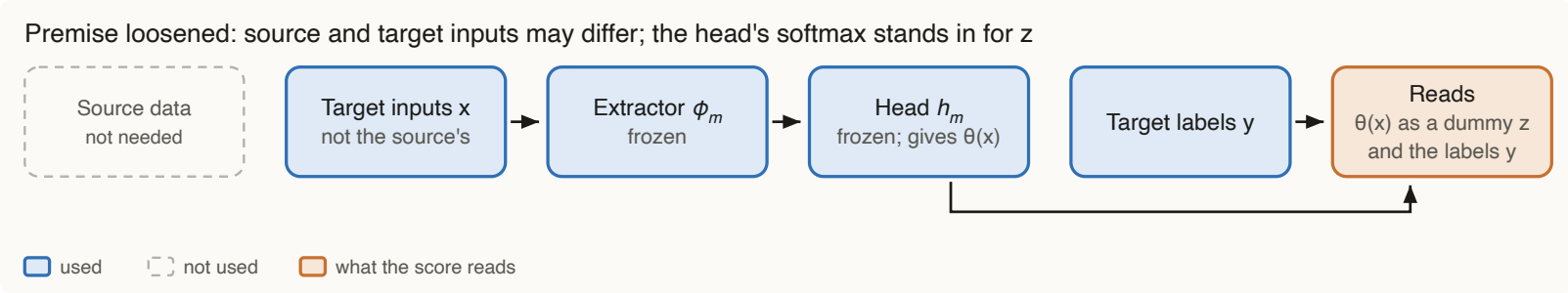


Two tasks on the **same inputs**, with ground-truth label sequences Z (source) and Y (target). No model, no features, no input data: count the joint $\hat{P}(y, z)$ and compute

$$H(Y \mid Z) = -\sum_{y,z} \hat{P}(y, z) \log \hat{P}(y \mid z)$$

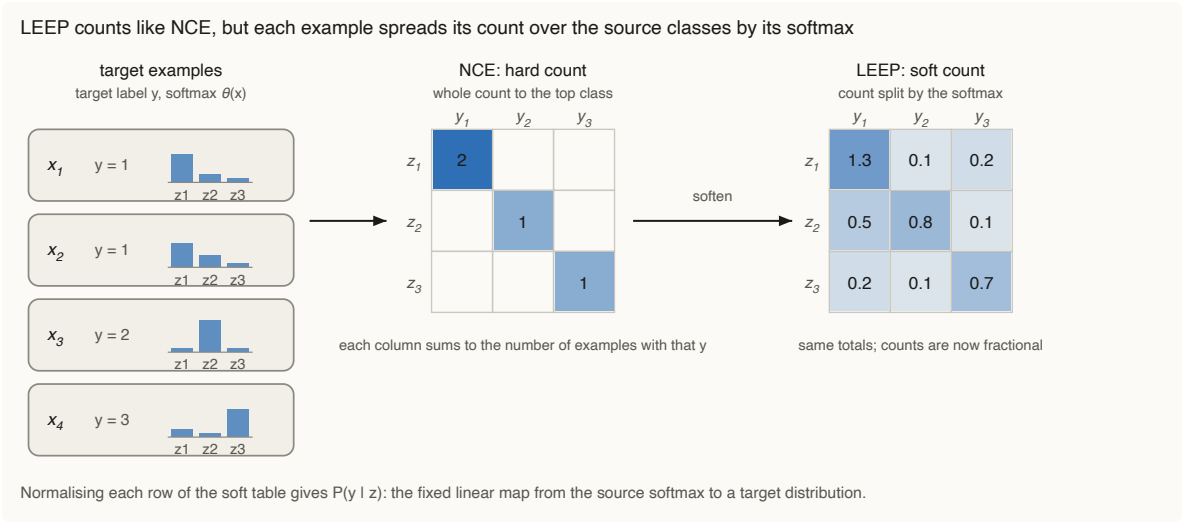


LEEP (Nguyen 2020): use the model's softmax instead



Push target inputs through the frozen source classifier: $\theta(x_i) \in \Delta(\mathcal{Z})$, a "dummy" label distribution, and count softly:

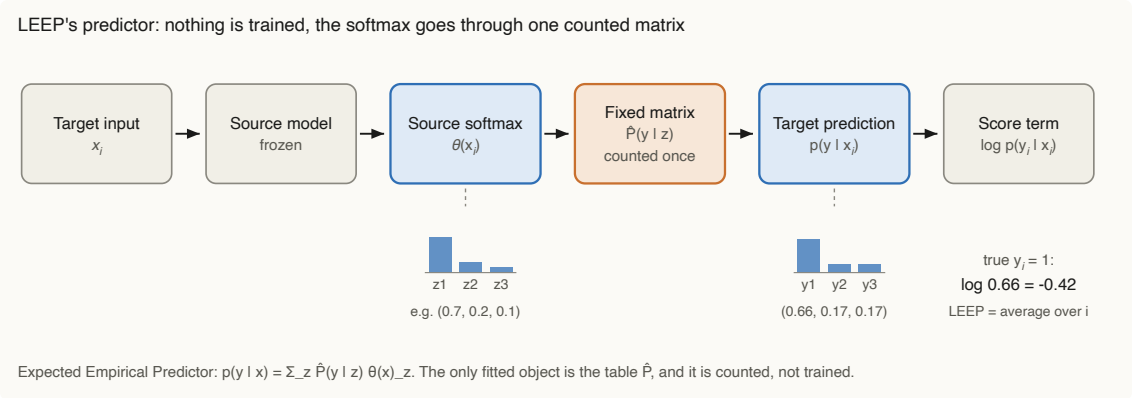
$$\hat{P}(y, z) = \frac{1}{n} \sum_{i:\, y_i = y} \theta(x_i)_z, \qquad \hat{P}(y \mid z) = \frac{\hat{P}(y, z)}{\hat{P}(z)}$$



LEEP: the Expected Empirical Predictor

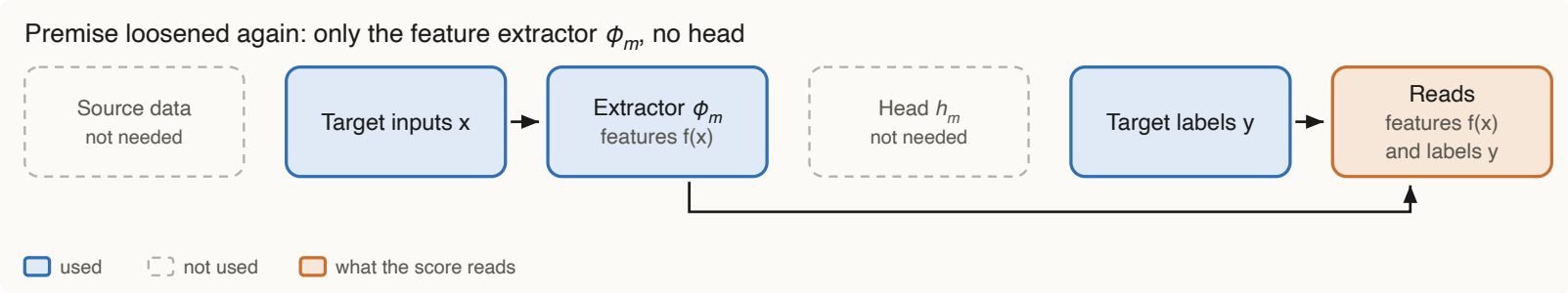
The **Expected Empirical Predictor** is a fixed linear map of the softmax, $p(\cdot \mid x) = \hat{P}^\top \theta(x)$, and

$$\mathrm{LEEP} = \frac{1}{n} \sum_{i=1}^{n} \log \sum_{z \in \mathcal{Z}} \hat{P}(y_i \mid z)\, \theta(x_i)_z$$



Notes: Nguyen 2020 — LEEP

H-score (Bao 2019): solve the head, then eliminate it



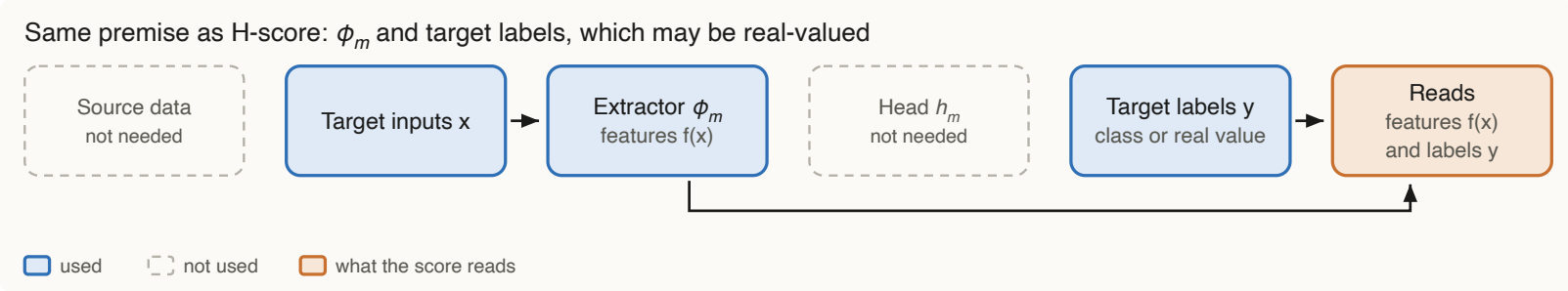
Frozen features $f(x) \in \mathbb{R}^k$, a linear head, target labels.

$$\mathcal{H}(f) = \operatorname{tr}\left(\operatorname{cov}\left(f(X)\right)^{-1} \operatorname{cov}\left(\mathbb{E}[f(X) \mid Y]\right)\right)$$



Notes: Bao 2019 — H-score

LogME (You 2021): integrate the head out



A Bayesian linear model on frozen features $F \in \mathbb{R}^{n \times D}$. Score the **evidence**, not the likelihood at the best w^*:

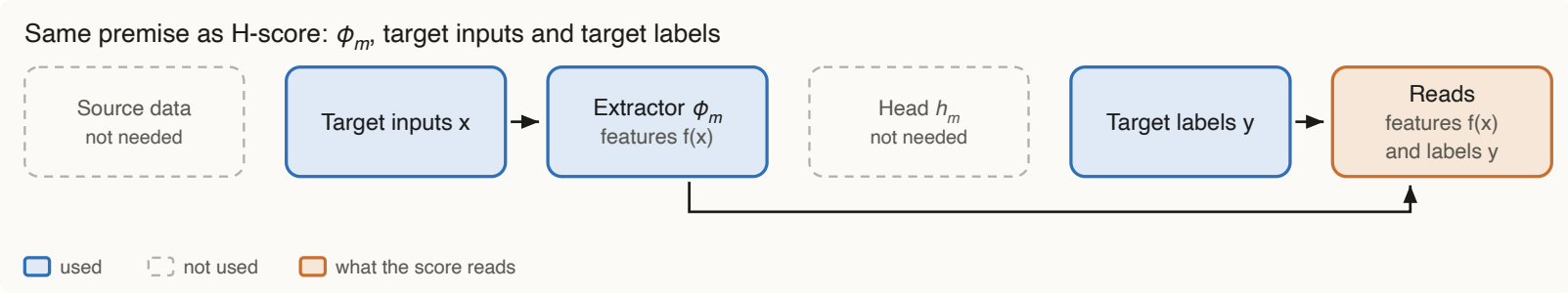
$$p(y \mid F, \alpha, \beta) = \int p(w \mid \alpha)\, p(y \mid F, w, \beta)\, \mathrm{d}w$$

$$\mathrm{LogME} = \frac{1}{n}\left[\frac{n}{2}\ln\beta^* + \frac{D}{2}\ln\alpha^* - \frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|A| - \frac{\beta^*}{2}\|Fm - y\|^2 - \frac{\alpha^*}{2}m^\top m\right]$$

with $A = \alpha^* I + \beta^* F^\top F$, $m = \beta^* A^{-1} F^\top y$, and α^*, β^* from a fixed point.

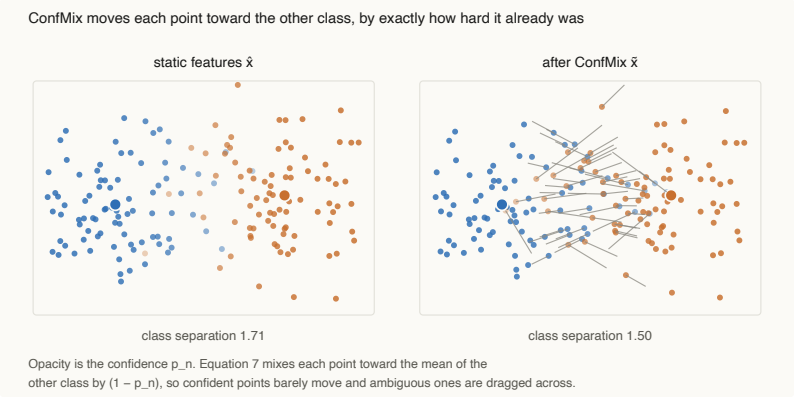
Notes: You 2021 — LogME

SFDA (Shao 2022): imitate fine-tuning, not only the features



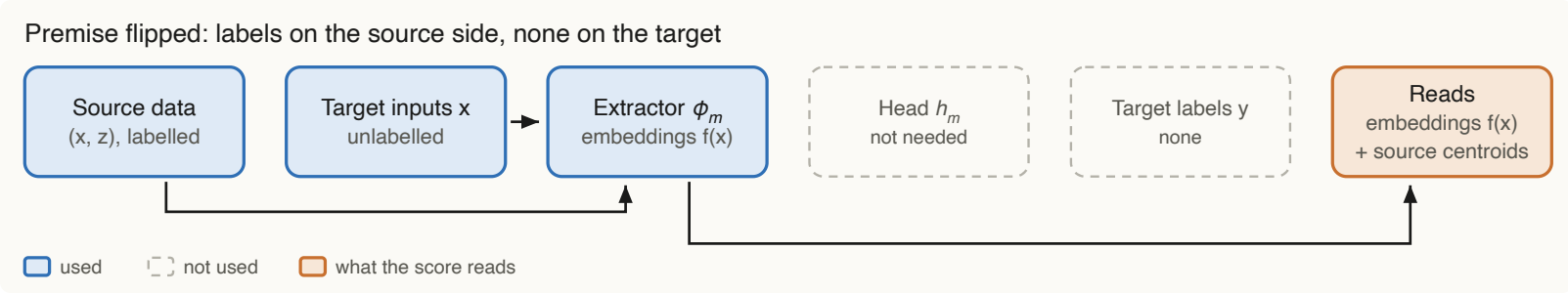
Fine-tuning separates classes and learns hard examples late, so a static score should imitate both.

- **Reg-FDA:** a regularised Fisher discriminant projects into $C - 1$ dimensions, LDA classifies there, each sample gets a confidence p_n
- **ConfMix:** moves each sample toward the other classes by its own difficulty. Re-run Reg-FDA; the score is the average log-likelihood of the second pass



Notes: Shao 2022 — SFDA

PAS (Diniz 2026): no target labels at all



Labelled source, unlabelled target. Unit-norm embeddings, source class centroids $\mu_c = \dfrac{\sum f_\theta(x_i^S)}{\left\|\sum f_\theta(x_i^S)\right\|}$, cosine distance $1 - f_\theta(x) \cdot \mu_c$. With d_1, d_2 the nearest and second-nearest centroid distances,

$$\mathrm{PAS} = \frac{1}{|\mathcal{D}^T|} \sum_j \frac{d_{2j} - d_{1j}}{d_{2j}} = 1 - \mathbb{E}\left[\frac{d_1}{d_2}\right]$$

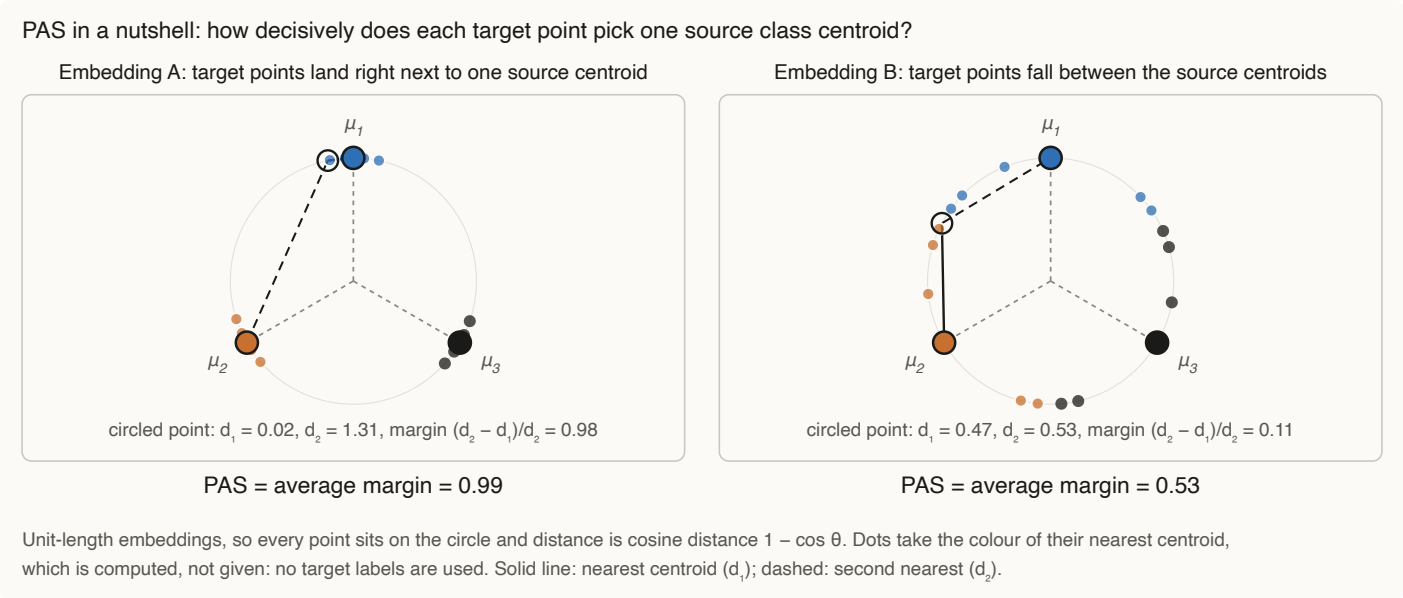
- A Silhouette score with "own cluster" replaced by "nearest cluster": the **average relative margin** of a nearest-centroid classifier, the complement of Lowe's ratio test
- Asymmetric by design. Range $[0, 1]$

Notes: Diniz 2026 — PAS

PAS: the core idea

PAS uses the **source** classes as a ruler and asks how decisively each unlabelled target point picks one of them:

1. embed to unit length and average each source class into a centroid μ_c
2. for each target point take the cosine distances to all centroids; d_1 and d_2 are the smallest and second smallest
3. its margin is $(d_2 - d_1)/d_2$: 1 on a centroid, 0 when equally close to two. PAS is the average margin

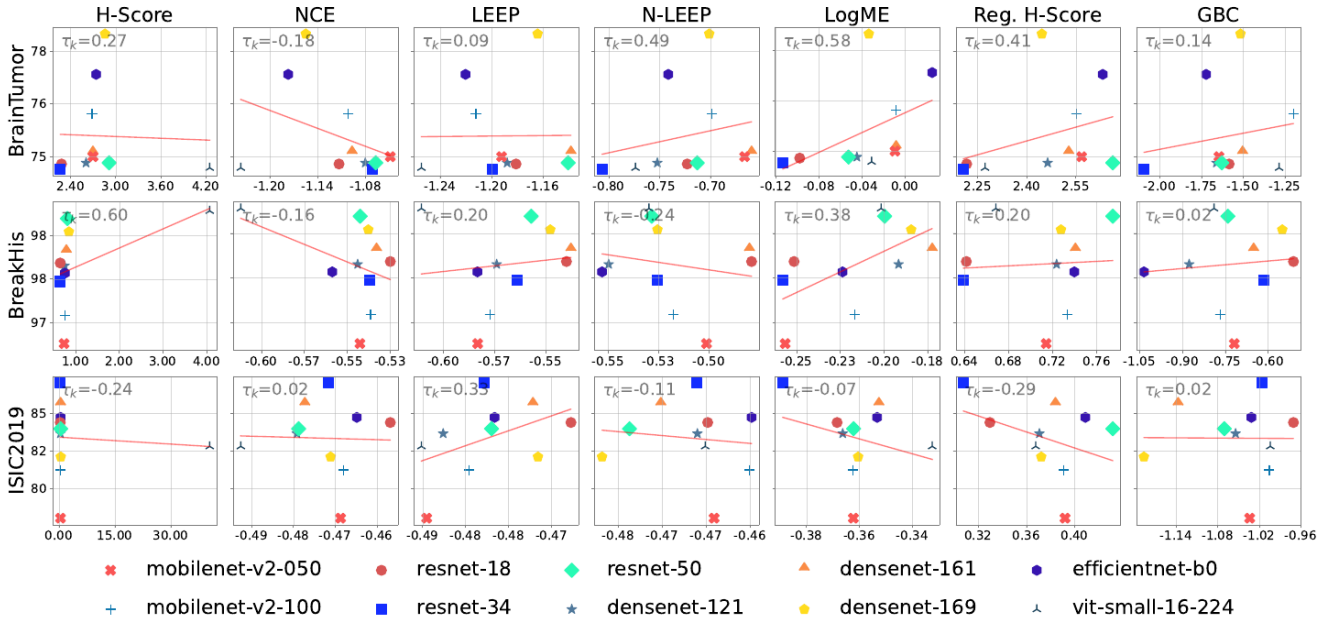


Notes: Diniz 2026 — PAS

Gaps

Do these scores survive medical imaging?

Chaves 2023: seven scores, ten models, three medical tasks



- **In-distribution:** unstable

Figure: Chaves et al. 2023, Fig. 3 (arXiv:2308.07444). Notes: Chaves 2023 — medical TE

Claßen 2026: do rankings survive a change of seed?

targets, sources	8 MedMNIST 2D sets; the other MedMNIST sets plus ImageNet
model	ResNet-18 only
metrics	H-score, LEEP, $\mathcal{N}$LEEP, LogME, NCTI, PARC, SFDA
design	nested stratified subsets of 5–75%, 5 seeds; reference tuned for accuracy and for AUROC

1. **Rankings move under nothing but the seed.** On Breast LogME ranks Blood as both the best and the worst source
2. **The reference ranking moves too.** Accuracy against AUROC reorders almost every source; one goes from 2nd to 11th

Also: $\mathcal{N}$LEEP and SFDA produce NaNs.

Notes: Claßen 2026 — TE robustness

Explorations

Mode connectivity as a transferability score

To do: mode connectivity between source and target

Question. Can the loss landscape between the source solution and the fine-tuned target solution explain, and predict, transferability?

- **Endpoints:** the source model $\theta_S = \theta_m$ from the pool, and the fine-tuned target model $\theta_T^* = \mathrm{FT}(\theta_m; \mathcal{D}^T)$, the **oracle** that defines T_m
- **Landscape:** the target loss $\mathcal{L}_T(\theta)$, and the source loss $\mathcal{L}_S(\theta)$ for contrast
- **Path:** $\gamma : [0, 1] \to \Theta$ with $\gamma(0) = \theta_S$ and $\gamma(1) = \theta_T^*$: a straight line, or a learned curve (Bézier or polygonal chain) that keeps the loss low, as in the mode-connectivity literature
- **Model:** fine-tuning as motion through this landscape. The height, length and bending of the path say how far the source solution must be reshaped for the target task

Related notes: Achille 2019 — task reachability (fine-tuning as a dynamical process)

To do: a path score as a transferability metric

Characterise the path by functionals of $\mathcal{L}_T$ along it:

$$\mathcal{I}_T = \int_0^1 \mathcal{L}_T\big(\gamma(t)\big)\,\|\dot\gamma(t)\|\,\mathrm{d}t, \qquad B = \max_t \Big[\mathcal{L}_T\big(\gamma(t)\big) - (1-t)\mathcal{L}_T(\theta_S) - t\,\mathcal{L}_T(\theta_T^*)\Big], \qquad \ell = \int_0^1 \|\dot\gamma(t)\|\,\mathrm{d}t$$

1. **Oracle runs:** fine-tune θ_T^* per source model, trace γ and record $\mathcal{I}_T$, B, ℓ and the curvature (Hessian or Fisher along the path)
2. **Explain:** which functional tracks the ground truth T_m, by τ_w and regret, against H-score, LogME, SFDA and the rest?
3. **Decode:** find a cheap proxy $\hat{S}_m$ for the winning functional that needs no fine-tuning (from features, Fisher spectrum or a few gradient steps), and calibrate it to the oracle
4. **Test:** rankings stable across seeds (Claßen 2026) and on the medical benchmarks (Chaves 2023)?

Open: a pre-trained and a fine-tuned model often share a basin, so the barrier may be near zero and the signal may sit in path length and curvature; permutation symmetry needs care when the endpoints differ more.

Sources

Notes (rendered): `msrepo.github.io/theory_inclined_papers_with_annotations/` followed by

`2019-tran-nce-hardness` · `2020-nguyen-leep` · `2019-bao-hscore-transferability` · `2021-you-logme` · `2022-shao-sfda` · `2026-diniz-pas` · `2023-chaves-medical-transferability` · `2026-classen-te-robustness`

and for the other axes: `2026-wang-ntk-selector`

Papers: Tran, Nguyen & Hassner, ICCV 2019, arXiv:1908.08142 · Nguyen et al., ICML 2020, arXiv:2002.12462 · Bao et al., ICIP 2019, arXiv:2212.10082 · You et al., ICML 2021, arXiv:2102.11005 · Shao et al., ECCV 2022, arXiv:2207.03036 · Diniz et al., ICLR 2026, arXiv:2604.09863 · Chaves et al., arXiv:2308.07444 · Claßen et al., arXiv:2608.09999